6.1 Equation of a Line in Slope y-intercept Form

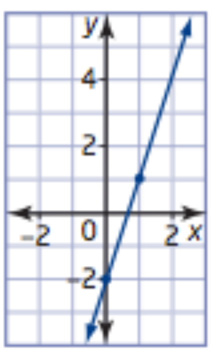
MPM1D

1. Identify the slope and the y-intercept of each line:

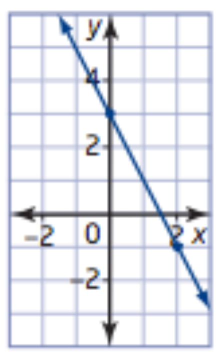
Equation	Slope	y-intercept
a) $y = 4x + 1$		
b) $y = \frac{2}{3}x + 3$		
c) $y = x - 2$		
d) $y = -\frac{2}{3}x$		
e) $y = 3$		
f) $y = -x - \frac{1}{2}$		

2. Find the slope and y-intercept of each line:

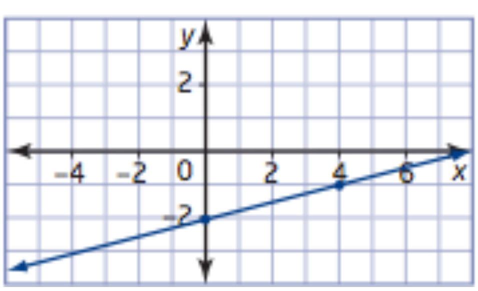
a)



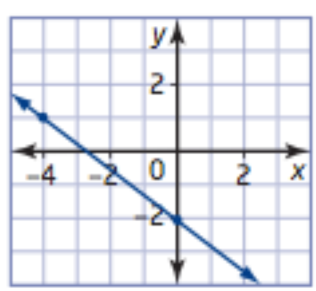
b)



c)



d)



3. Write the equation of each line in question #2

a)

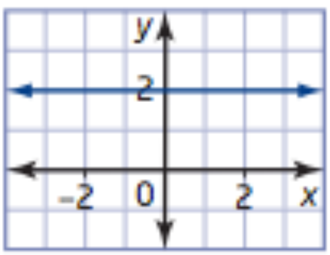
b)

c)

d)

4. Write the equation of each line. State its slope and y-intercept if they exist:

a)

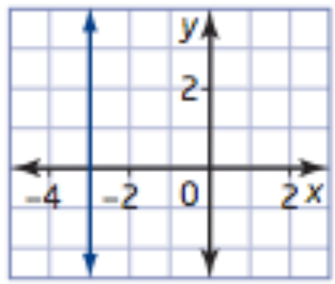


Equation:

Slope:

y-intercept:

b)

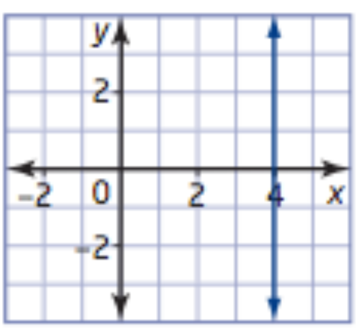


Equation:

Slope:

y-intercept:

c)

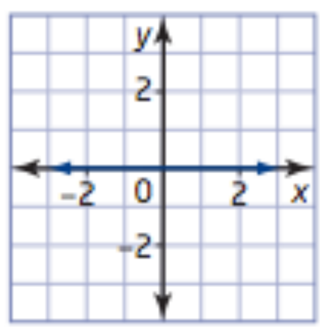


Equation:

Slope:

y-intercept:

d)



Equation:

Slope:

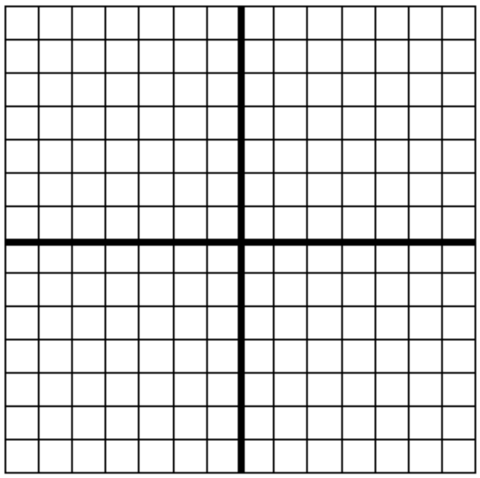
y-intercept:

5. The line in question 4 part d) has a special name. What is it?

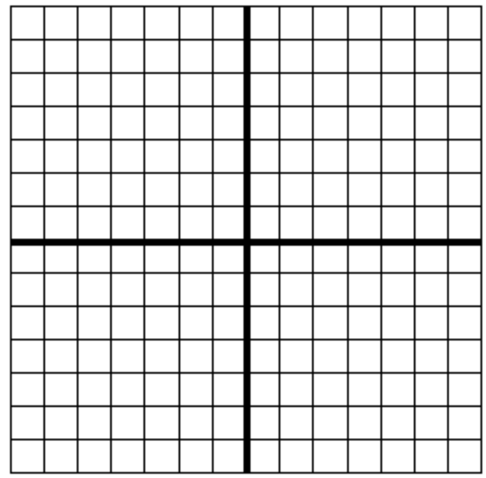
6. The slope and y-intercept are given. Write the equation and graph each line:

	Slope	y-intercept
a)	$\frac{2}{3}$	3
b)	$-\frac{3}{5}$	1
c)	−2	0
d)	$\frac{4}{3}$	−4
e)	0	−4

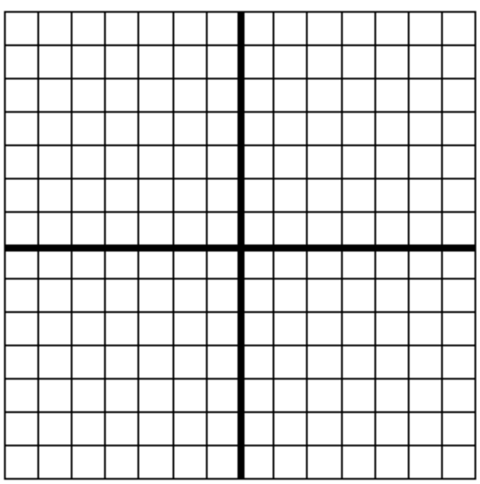
a) Equation:



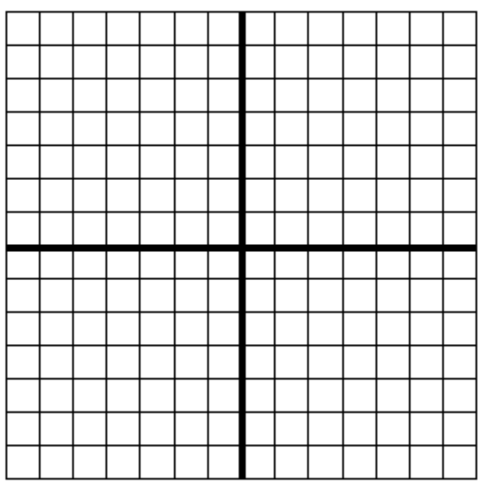
b) Equation:



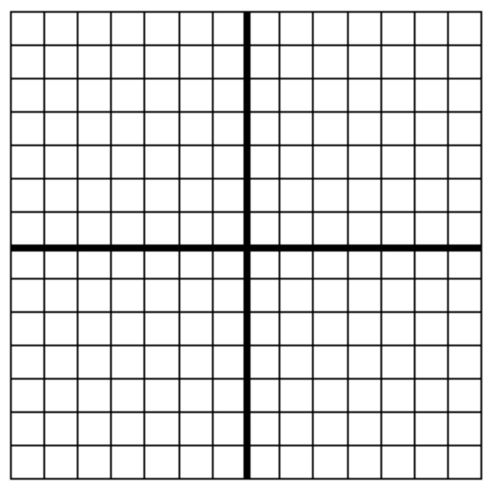
c) Equation:



d) Equation:



e) Equation

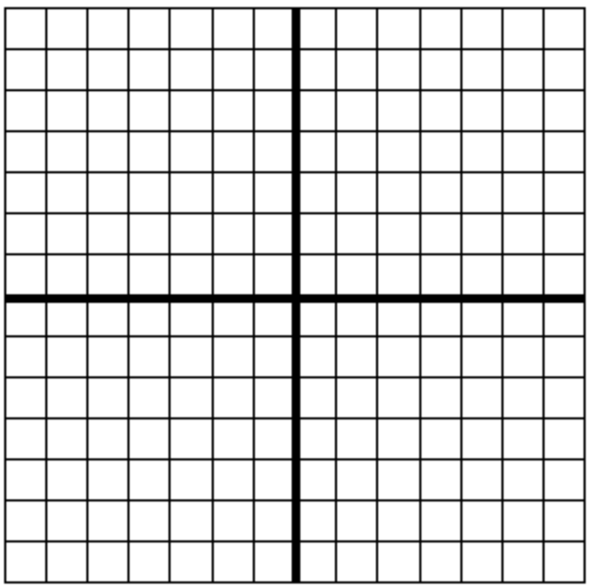


7. State the slope and y-intercept of each line, if they exist. Graph each line.

a) $y = -5$

slope:

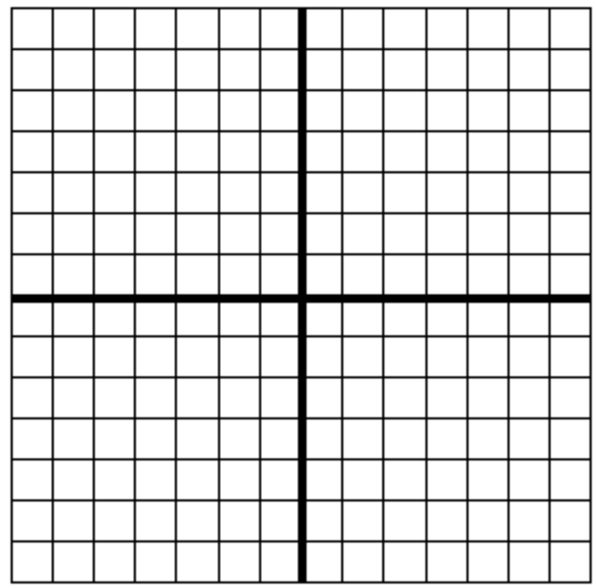
y-intercept:



b) $x = 1$

slope:

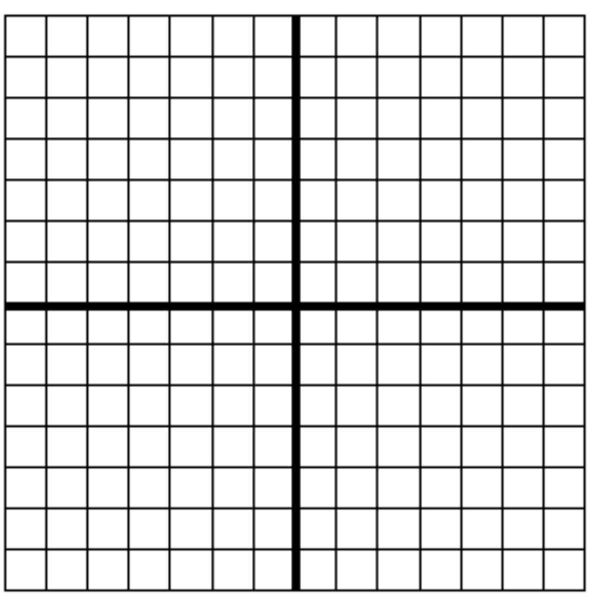
y-intercept:



c) $y = \frac{7}{2}$

slope:

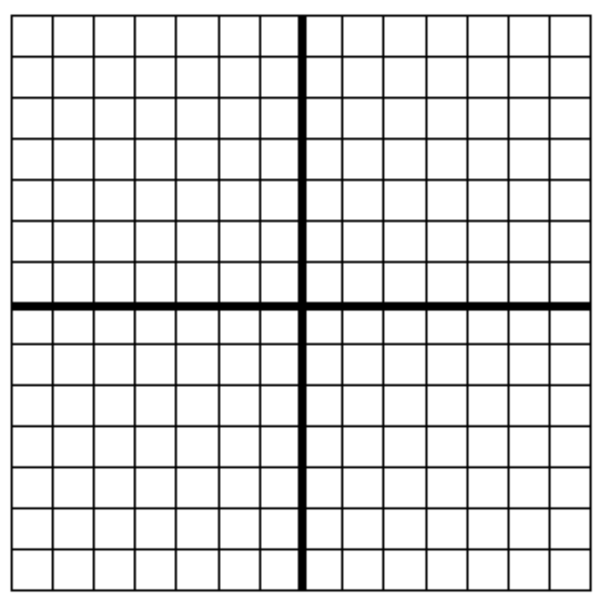
y-intercept:



b) $x = -2.5$

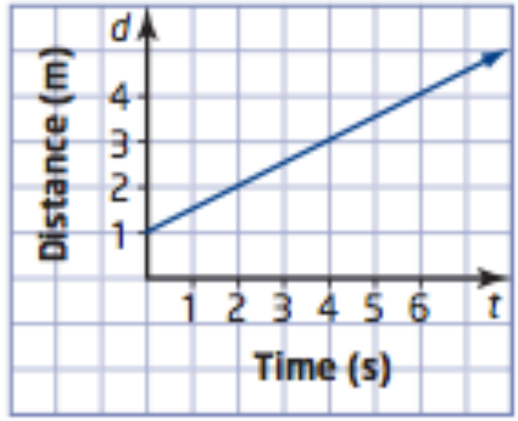
slope:

y-intercept:



8. The distance-time graph of a person walking in front to a motion sensor is shown.

a) How far from the sensor did the person begin walking?

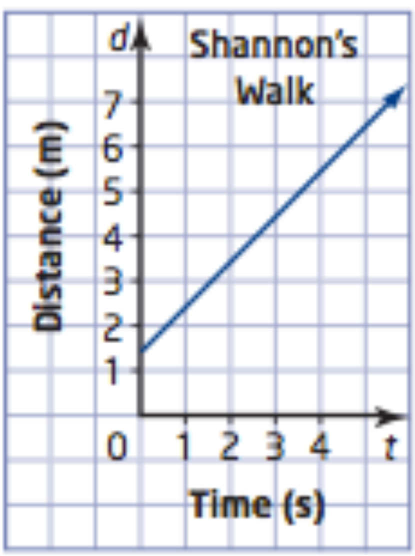


b) How fast did the person walk? (find the slope)

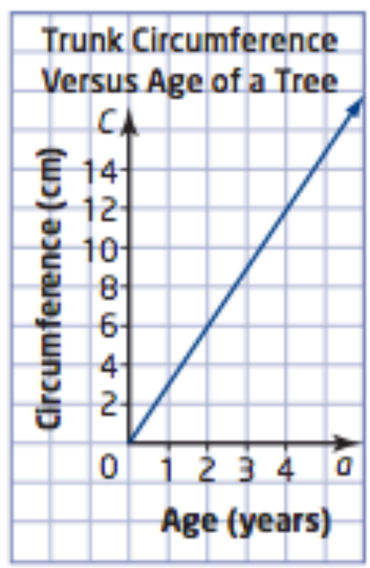
c) Did the person walk away from or toward the sensor? Explain.

10. Identify the slope and the vertical intercept of each linear relation and explain what they represent. Write an equation to describe the relationship.

a)



b)



Slope:

y-intercept:

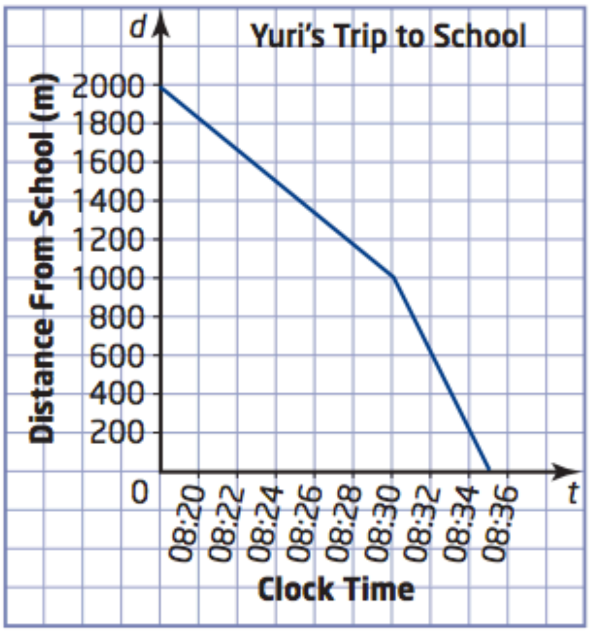
Equation:

Slope:

y-intercept:

Equation:

12. Yuri tries hard not to be late for class, but sometimes he does not quite make it on time. Class begins at 8:30 A.M. The distance-time graph shows his progress from home to school one morning.



Write a story about Yuri's trip to school. Include the speed, distance, and time in your story.